

4. Raspberry Pi Development

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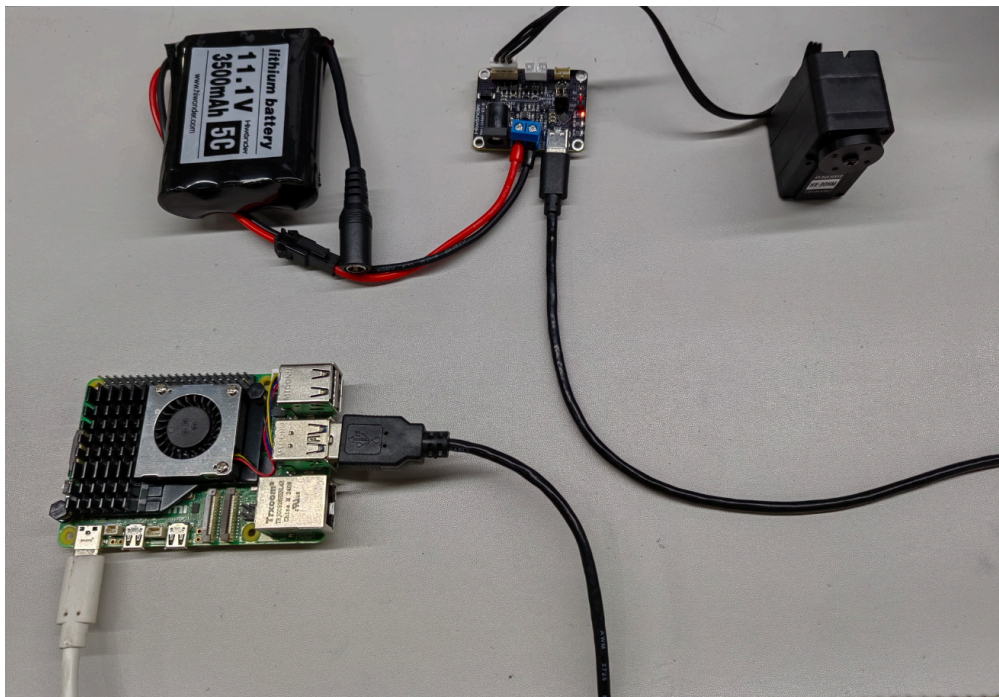
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4.1. Getting Started

4.1.1 Wiring Instruction

This example uses a Raspberry Pi 5 controller powered by a 5 V adapter. The bus servo is connected to the debugging board, which is powered by an 11.1 V lithium battery. Finally, the debugging board is connected to any USB port of the Raspberry Pi 5 using a Type-C cable.



Note

- When using Hiwonder's lithium battery, connect the battery cable with the red wire to the positive (+) terminal and the black wire to the negative (-) terminal of the DC port.

- Before connecting the battery cables, make sure they are not already attached to the lithium battery. This prevents the risk of a short circuit caused by accidental contact between the positive and negative wires.

4.1.2 Environment Configuration

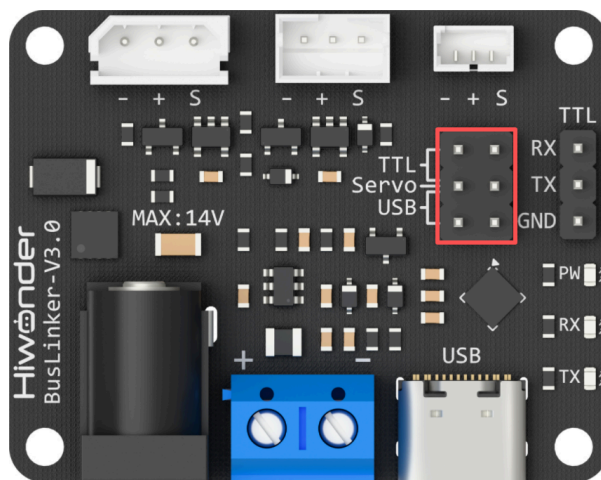
Install VNC software on PC. The software package is stored in [2. Softwares \ Remote Connection Software](#). For the detailed operations of VNC, please refer to the relevant tutorials.

Drag the program and SDK library files into the Raspberry Pi system image. In this example, they are placed in the Demo folder on the desktop. Note: Make sure the library files are placed in the same directory as the program.

4.2 Development Example

Note

Before running this example, ensure that the jumpers on the debugging board are installed on the Servo and USB pins. Otherwise, communication will not function properly.



4.2.1 Reading the Servo Status

This example displays the bus servo status information through the terminal window.

4.2.1.1 Run Program

1. Open a new terminal and enter the following command to navigate to the program directory:

```
cd Desktop/Demo/examples
```

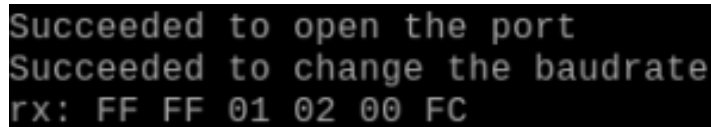


2. Run the program by entering:

```
python3 ping.py
```

A terminal window with a black background. The text 'python3 ping.py' is written in a green monospace font. A white cursor is positioned at the end of the command.

4.2.1.2 Program Outcome

A terminal window with a black background. The text is displayed in a light gray monospace font. It shows three lines of output: 'Succeeded to open the port', 'Succeeded to change the baudrate', and 'rx: FF FF 01 02 00 FC'.

After running the program, the terminal prints the servo status information, which corresponds to the data frames returned by the servo.

4.2.1.3 Program Brief Analysis

- Import the required libraries.

```
import sys
import os

sys.path.append("../")

from servo_sdk import *
```

- Initialize the communication port and servo objects.

```
PortHandler = PortHandler('/dev/ttyACM0')
ServoHandler = HxServoHandler(PortHandler)
```

- Create and open a port object, then verify and set the port baud rate.

```
if PortHandler.openPort():
    print("Succeeded to open the port")
else:
    print("Failed to open the port")
    quit()

if PortHandler.setBaudRate(1000000):
    print("Succeeded to change the baudrate")
else:
    print("Failed to change the baudrate")
    quit()
```

- Read the operating status of servo ID 1 and print the response packets returned by the servo.

```
rxpacket, result, error = ServoHandler.ping(1)
if result != COMM_SUCCESS:
    print("%s" % ServoHandler.getTxRxResult(result))
else:
    print(f"rx: {' '.join(f'{byte:02X}' for byte in rxpacket)}")
if error != 0:
    print("%s" % ServoHandler.getRxPacketError(error))
```

4.2.2 Reading the Servo Position

This example displays the bus servo position information through the terminal window.

4.2.2.1 Run Program

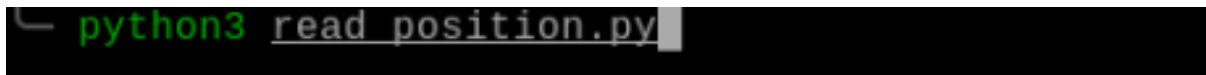
1. Open a new terminal and enter the following command to navigate to the program directory:

```
cd Desktop/Demo/examples
```



2. Run the program by entering:

```
python3 read_position.py
```



4.2.2.2 Program Outcome

```
Succeeded to open the port
Succeeded to change the baudrate
550
550
550
550
550
550
550
550
550
550
550
```

After running the program, the terminal continuously prints the current position information of the servo.

4.2.2.3 Program Brief Analysis

- Import the required libraries.

```
import sys
import os

sys.path.append("..")

from servo_sdk import *
```

- Initialize the communication port and servo objects.

```
PortHandler = PortHandler('/dev/ttyACM0')
ServoHandler = HxServoHandler(PortHandler)
```

- Create and open a port object, then verify and set the port baud rate.

```
if PortHandler.openPort():
    print("Succeeded to open the port")
else:
    print("Failed to open the port")
    quit()

if PortHandler.setBaudRate(1000000):
    print("Succeeded to change the baudrate")
else:
    print("Failed to change the baudrate")
    quit()
```

- The `readLoad` function continuously reads the current position of servo ID 1 and prints it, then closes the port.

```
while True:
    pos, result, error = ServoHandler.readLoad(1)
    if result != COMM_SUCCESS:
        print("%s" % ServoHandler.getTxRxResult(result))
        break

    if error != 0:
        print("%s" % ServoHandler.getRxPacketError(error))
        break

    print(pos)
    time.sleep(0.1) # Delay for 0.1 seconds

PortHandler.closePort()
```

4.2.3 Servo Movement Control in Write Mode

4.2.3.1 Run Program

1. Open a new terminal and enter the following command to navigate to the program directory:

```
cd Desktop/Demo/examples
```



2. Run the program by entering:

```
python3 write_position.py
```



4.2.3.2 Program Outcome

After running the program, servo ID 1 moves to position -3000 at a speed of 100 steps per second and an acceleration of 0 steps per second squared.

4.2.3.3 Program Brief Analysis

- Import the required libraries.

```
import sys
import os

sys.path.append("..")

from servo_sdk import *
```

- Initialize the communication port and servo objects.

```
PortHandler = PortHandler('/dev/ttyACM0')
ServoHandler = HxServoHandler(PortHandler)
```

- Create and open a port object, then verify and set the port baud rate.

```

if PortHandler.openPort():
    print("Succeeded to open the port")
else:
    print("Failed to open the port")
    quit()

if PortHandler.setBaudRate(1000000):
    print("Succeeded to change the baudrate")
else:
    print("Failed to change the baudrate")
    quit()

```

- The `writePosEx` function is used to control the servo movement, and the port is closed afterward.

```

result, error = ServoHandler.writePosEx(1, -3000, 100, 0)

if result != COMM_SUCCESS:
    print("%s" % ServoHandler.getTxRxResult(result))
if error != 0:
    print("%s" % ServoHandler.getRxPacketError(error))

# Close port
PortHandler.closePort()

```

4.2.4 Servo Movement Control in RegWrite Mode

4.2.4.1 Run Program

1. Open a new terminal and enter the following command to navigate to the program directory:

```
cd Desktop/Demo/examples
```



2. Run the program by entering:

```
python3 write_position.py
```



4.2.4.2 Program Outcome

After running the program, servo ID 1 moves back and forth between positions 20 and 1000 at a speed of 1500 steps per second and an acceleration of 0 steps per second squared, with an interval of approximately 0.75 seconds.

4.2.4.3 Program Brief Analysis

- Import the required libraries.

```
import sys
import os

sys.path.append("..")

from servo_sdk import *
```

- Initialize the communication port and servo objects.

```
PortHandler = PortHandler('/dev/ttyACM0')
ServoHandler = HxServoHandler(PortHandler)
```

- Create and open a port object, then verify and set the port baud rate.

```
if PortHandler.openPort():
    print("Succeeded to open the port")
else:
    print("Failed to open the port")
    quit()

if PortHandler.setBaudRate(1000000):
    print("Succeeded to change the baudrate")
else:
    print("Failed to change the baudrate")
    quit()
```

- The `writeRegPosEx` function is used to control the servo movement, and the `regAction` function triggers the execution of `writeRegPosEx`. The port is closed afterward.

```
while True:
    result, error = ServoHandler.writeRegPosEx(1, 1000, 1500, 0)
    if result != COMM_SUCCESS:
        print("%s" % ServoHandler.getTxRxResult(result))
        break

    if error != 0:
        print("%s" % ServoHandler.getRxPacketError(error))
        break
    ServoHandler.regAction()
    time.sleep(((1000-20) / 1500) + 0.1)  #//[(P1-P0)/(V)] + 0.1

    # Servo (ID1~10) runs at a maximum speed of v=1500*0.059=88.5rpm until it reaches
    position P0=20
    result, error = ServoHandler.writeRegPosEx(1, 20, 1500, 0)
    if result != COMM_SUCCESS:
        print("%s" % ServoHandler.getTxRxResult(result))
        break
```



```

if error != 0:
    print("%s" % ServoHandler.getRxPacketError(error))
    break

ServoHandler.regAction()
time.sleep(((1000 - 20) / 1500) + 0.1)
# time.sleep((1000-20)/(1500) + 0.1)) #//[(P1-P0)/(V)] + 0.1

# Close port
PortHandler.closePort()

```

4.2.5 Servo Movement Control in SyncWrite Mode

4.2.5.1 Run Program

1. Open a new terminal and enter the following command to navigate to the program directory:

```
cd Desktop/Demo/examples
```



2. Run the program by entering:

```
python3 sync_write.py
```



4.2.5.2 Program Outcome

After running the program, servos ID 1 to 3 simultaneously move back and forth between positions 20 and 1000 at a speed of 1500 steps per second and an acceleration of 0 steps per second squared, with an interval of approximately 0.75 seconds.

4.2.5.3 Program Brief Analysis

- Import the required libraries.

```

import sys
import os

sys.path.append("..")

from servo_sdk import *

```

- Initialize the communication port and servo objects.

```
PortHandler = PortHandler('/dev/ttyACM0')
ServoHandler = HxServoHandler(PortHandler)
```

- Create and open a port object, then verify and set the port baud rate.

```
if PortHandler.openPort():
    print("Succeeded to open the port")
else:
    print("Failed to open the port")
    quit()

if PortHandler.setBaudRate(1000000):
    print("Succeeded to change the baudrate")
else:
    print("Failed to change the baudrate")
    quit()
```

- The `syncwritePosEx` function is used to control the servo movement, and the port is closed afterward.

```
while True:
    for id in range(1, 3):
        add_param_result = ServoHandler.syncwritePosEx(id, 1000, 1500, 0)
        if add_param_result != True:
            print("[ID:%03d] groupSyncwrite addparam failed" % id)

    # Syncwrite goal position
    result = ServoHandler.GroupSyncwrite.txPacket()
    if result != COMM_SUCCESS:
        print("%s" % result.getTxRxResult(result))
        break

    time.sleep(((1000-20) / 1500) + 0.1) #//[(P1-P0)/(V)] + 0.1

    # Clear syncwrite parameter storage
    ServoHandler.GroupSyncwrite.clearParam()

    for id in range(1, 3):
        add_param_result = ServoHandler.syncwritePosEx(id, 20, 1500, 0)
        if add_param_result != True:
            print("[ID:%03d] groupSyncwrite addparam failed" % id)

    # Syncwrite goal position
    result = ServoHandler.GroupSyncwrite.txPacket()
    if result != COMM_SUCCESS:
        print("%s" % result.getTxRxResult(result))
        break

    time.sleep(((1000 - 20) / 1500) + 0.1)
    # time.sleep(((1000-20)/(1500) + 0.1)) #//[(P1-P0)/(V)] + 0.1

    # Clear syncwrite parameter storage
```

```
ServoHandler.GroupSyncWrite.clearParam()
# Close port
PortHandler.closePort()
```

4.2.6 Setting the Servo Operating Mode

4.2.6.1 Run Program

1. Open a new terminal and enter the following command to navigate to the program directory:

```
cd Desktop/Demo/examples
```

A terminal window with a dark background. The command 'cd Desktop/Demo/examples' is entered in green text. The cursor is at the end of the command.

2. Run the program by entering:

```
python3 select_mode.py
```

A terminal window with a dark background. The command 'python3 select_mode.py' is entered in green text. The cursor is at the end of the command.

4.2.6.2 Program Outcome

After running the program, servo ID 1 is set to Position mode.

4.2.6.3 Program Brief Analysis

- Import the required libraries.

```
import sys
import os

sys.path.append("..")

from servo_sdk import *
```

- Initialize the communication port and servo objects.

```
PortHandler = PortHandler('/dev/ttyACM0')
ServoHandler = HxServoHandler(PortHandler)
```

- Create and open a port object, then verify and set the port baud rate.

```

if PortHandler.openPort():
    print("Succeeded to open the port")
else:
    print("Failed to open the port")
    quit()

if PortHandler.setBaudRate(1000000):
    print("Succeeded to change the baudrate")
else:
    print("Failed to change the baudrate")
    quit()

```

- The `selectMode` function is used to set the servo operating mode.

```

rxpacket, result, error = ServoHandler.selectMode(1, 0)

if result != COMM_SUCCESS:
    print("%s" % ServoHandler.getTxRxResult(result))
else:
    print(f"rx: {' '.join(f'{byte:02X}' for byte in rxpacket)}")
if error != 0:
    print("%s" % ServoHandler.getRxPacketError(error))

```